NAGATA RINGS AND NOETHERIAN UNIVERSALLY JAPANESE RINGS ARE EQUIVALENT

THE STACKS PROJECT AUTHORS

Editorial note. Original author theorem and proof are retained. Only this target is counted; local supporting statements are not separate new entries.

Allowed standard prerequisites (editorial).

- Noetherian normalization, N-1 and N-2 criteria, polynomial N-2 stability, quasi-finite Nagata ascent, Hilbert Nullstellensatz and the cited standard analytic-unramified criteria.. These are the named standard inputs; the selected main construction and its local nonroutine support remain in full.

Source reference. Original lemma 032O: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4027–4027. Stable Stacks locator for declared standard input or surrounding context. No selected local proof is replaced by this pointer.

Source reference. Original lemma 032T: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4032–4032. Stable Stacks locator for declared standard input or surrounding context. No selected local proof is replaced by this pointer.

Source reference. Original lemma 0333: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4042–4042. Stable Stacks locator for declared standard input or surrounding context. No selected local proof is replaced by this pointer.

Source reference. Original theorem 00FV: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 585–585. Stable Stacks locator for declared standard input or surrounding context. No selected local proof is replaced by this pointer.

Source reference. Original lemma 032K: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4023–4023. Stable Stacks locator for declared standard input or surrounding context. No selected local proof is replaced by this pointer.

Source reference. Original lemma 0330: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4039–4039. Stable Stacks locator for declared standard input or surrounding context. No selected local proof is replaced by this pointer.

Source reference. Original lemma 032Y: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4037–4037. Stable Stacks locator for declared standard input or surrounding context. No selected local proof is replaced by this pointer.

Source reference. Original lemma 00PQ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 940–940. Stable Stacks locator for declared standard input or surrounding context. No selected local proof is replaced by this pointer.

Date: stacks-project:algebra.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original section 031T: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3996–3996. Stable Stacks locator for declared standard input or surrounding context. No selected local proof is replaced by this pointer.

Source reference. Original lemma 032U: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4033–4033. Stable Stacks locator for declared standard input or surrounding context. No selected local proof is replaced by this pointer.

Source reference. Original lemma 0331: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4040–4040. Stable Stacks locator for declared standard input or surrounding context. No selected local proof is replaced by this pointer.

Proposition 0.1 (Nagata)**.** *Let R be a ring. The following are equivalent:*

(1) *R is a Nagata ring,*
(2) *any finite type R-algebra is Nagata, and*
(3) *R is universally Japanese and Noetherian.*

Proof. It is clear that a Noetherian universally Japanese ring is universally Nagata (i.e., condition (2) holds). Let R be a Nagata ring. We will show that any finitely generated R-algebra S is Nagata. This will prove the proposition.

Step 1. There exists a sequence of ring maps $R = R_0 \to R_1 \to R_2 \to \ldots \to R_n = S$ such that each $R_i \to R_{i+1}$ is generated by a single element. Hence by induction it suffices to prove S is Nagata if $S \cong R[x]/I$.

Step 2. Let $\mathfrak{q} \subset S$ be a prime of S, and let $\mathfrak{p} \subset R$ be the corresponding prime of R. We have to show that $S/\mathfrak{q}$ is N-2. Hence we have reduced to proving the following: (*) Given a Nagata domain R and a monogenic extension $R \subset S$ of domains then S is N-2.

Step 3: Let R be a Nagata domain and let $R \subset S$ be a monogenic extension of domains. Suppose the induced extension of fraction fields of R and S is purely transcendental. In this case $S = R[x]$. By Lemma 032O we see that S is N-2. Hence we have reduced to proving the following: (**) Given a Nagata domain R and a monogenic extension $R \subset S$ of domains inducing a finite extension of fraction fields then S is N-2.

Step 4. Let R be a normal Nagata domain and let $R \subset S$ be a monogenic extension of domains inducing a finite extension of fraction fields L/K. Choose an element $x \in S$ which generates S as an R-algebra. Let M/L be a finite extension of fields. Let R' be the integral closure of R in M. Then the integral closure S' of S in M is equal to the integral closure of $R'[x]$ in M. Also the fraction field of R' is M and $R \subset R'$ is finite (by the Nagata property of R). This implies that R' is a Nagata ring (Lemma 032T). To show that S' is finite over S is the same as showing that S' is finite over $R'[x]$. Replace R by R' and S by $R'[x]$ to reduce to the following statement: (***) Given a normal Nagata domain R with fraction field K, and $x \in K$, the ring $S \subset K$ generated by R and x is N-1.

Step 5. Let R be a normal Nagata domain with fraction field K. Let $x = b/a \in K$. We have to show that the ring $S \subset K$ generated by R and x is N-1. Note that $S_a \cong R_a$ is normal. Hence by Lemma 0333 it suffices to show that $S_{\mathfrak{m}}$ is N-1 for every maximal ideal $\mathfrak{m}$ of S.

With assumptions as in the preceding paragraph, pick such a maximal ideal and set $\mathfrak{n} = R \cap \mathfrak{m}$. The residue field extension $\kappa(\mathfrak{m})/\kappa(\mathfrak{n})$ is finite (Theorem 00FV) and

generated by the image of x. After replacing R by R_a for some $a \in R$, $a \notin \mathfrak{n}$ and S by S_a, may assume there exists a monic polynomial $f(X) = X^d + \sum_{i=1,\ldots,d} a_i X^{d-i}$ in $R[x]$ with $f(x) \in \mathfrak{m}$ (details omitted). Let K''/K be a finite extension of fields such that the polynomial $f(X)$ splits completely in $K''[X]$. Let R' be the integral closure of R in K''. Let $S' \subset K''$ be the subring generated by R' and x. As R is Nagata we see R' is finite over R and Nagata (Lemma 032T). Moreover, S' is finite over S. If for every maximal ideal $\mathfrak{m}'$ of S' the local ring $S'_{\mathfrak{m}'}$ is N-1, then $S'_{\mathfrak{m}}$ is N-1 by Lemma 0333, which in turn implies that $S_{\mathfrak{m}}$ is N-1 by Lemma 032K. After replacing R by R' and S by S', and $\mathfrak{m}$ by any of the maximal ideals $\mathfrak{m}'$ lying over $\mathfrak{m}$ we reach the situation where the polynomial f above split completely: $f(X) = \prod_{i=1,\ldots,d}(X - a_i)$ with $a_i \in R$. Since $f(x) \in \mathfrak{m}$ we see that $x - a_i \in \mathfrak{m}$ for some i. Finally, after replacing x by $x - a_i$ we may assume that $x \in \mathfrak{m}$.

To recapitulate: R is a normal Nagata domain with fraction field K, $x \in K$ and S is the subring of K generated by x and R, finally $\mathfrak{m} \subset S$ is a maximal ideal with $x \in \mathfrak{m}$. We have to show $S_{\mathfrak{m}}$ is N-1.

We will show that Lemma 0330 applies to the local ring $S_{\mathfrak{m}}$ and the element x. This will imply that $S_{\mathfrak{m}}$ is analytically unramified, whereupon we see that it is N-1 by Lemma 032Y.

We have to check properties (1), (2), (3)(a) and (3)(b). Property (1) is trivial. Let $I = \mathrm{Ker}(R[X] \to S)$ where $X \mapsto x$. We claim that I is generated by all linear forms $aX - b$ such that $ax = b$ in K. Clearly all these linear forms are in I. If $g = a_d X^d + \ldots a_1 X + a_0 \in I$, then we see that $a_d x$ is integral over R (Lemma 00PQ) and hence $b := a_d x \in R$ as R is normal. Then $g - (a_d X - b)X^{d-1} \in I$ and we win by induction on the degree. As a consequence we see that

$$S/xS = R[X]/(X, I) = R/J$$

where

$$J = \{b \in R \mid ax = b \text{ for some } a \in R\} = xR \cap R$$

By Lemma 031T we see that $S/xS = R/J$ has no embedded primes as an R-module, hence as an R/J-module, hence as an S/xS-module, hence as an S-module. This proves property (2). Take such an associated prime $\mathfrak{q} \subset S$ with the property $\mathfrak{q} \subset \mathfrak{m}$ (so that it is an associated prime of $S_{\mathfrak{m}}/xS_{\mathfrak{m}}$ – it does not matter for the arguments). Then $\mathfrak{q}$ is minimal over xS and hence has height 1. By the sequence of equalities above we see that $\mathfrak{p} = R \cap \mathfrak{q}$ is an associated prime of R/J, and so has height 1 (see Lemma 031T). Thus $R_{\mathfrak{p}}$ is a discrete valuation ring and therefore $R_{\mathfrak{p}} \subset S_{\mathfrak{q}}$ is an equality. This shows that $S_{\mathfrak{q}}$ is regular. This proves property (3)(a). Finally, $(S/\mathfrak{q})_{\mathfrak{m}}$ is a localization of $S/\mathfrak{q}$, which is a quotient of $S/xS = R/J$. Hence $(S/\mathfrak{q})_{\mathfrak{m}}$ is a localization of a quotient of the Nagata ring R, hence Nagata (Lemmas 032T and 032U) and hence analytically unramified (Lemma 0331). This shows (3)(b) holds and we are done. □